COM4569/CMP4569 DATA MINING

2026-2027 FALL SEMESTER

COURSE SYLLABUS & INTRODUCTION

Lecturer: Asst. Prof. Dr. Damla Topallı
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Office Hours (Office-M3):

Monday: 13:30- 15:00

Wednesday: 13.30-15:00

Course Information:

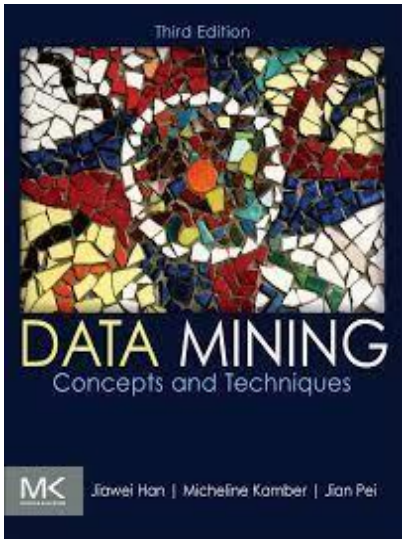
This course introduces the fundamental concepts, techniques, and applications of data mining. It focuses on extracting useful patterns and knowledge from data by combining concepts from databases, statistics, machine learning, and data analysis. Students will study the complete data mining process, including data understanding, preprocessing, exploratory analysis, pattern discovery, predictive modeling, clustering, and evaluation. Data mining is best learned by **working with data**. The course will combine theoretical foundations with practical exercises and applications using real-world datasets.

Weekly Schedule:

1	Introduction to Data Mining
2	Fundamental Concepts of Data Mining
3	Data Preprocessing Techniques
4	Data Warehouses and OLAP
5	Feature Extraction – Feature Selection
6	Classification Algorithms - Linear Models - Naive Bayes, k-Nearest Neighbor (kNN) Algorithm
7	Classification Algorithms - Rule-Based Algorithms, Decision Trees
8	Midterm Exam (Tentative; the exact date will be announced according to the Department's examination schedule.)
9	Classification Algorithms - Support Vector Machines (SVM)
10	Measuring Model Performance - Performance Metrics
11	Clustering Algorithms - Mathematical Modeling, k-Means
12	Clustering Algorithms - Hierarchical Clustering and Density-Based Clustering
13	Data Mining Applications - Text Mining
14	Data Mining Applications - Web Mining
15	Project Presentations & Demo
16	Course Review and Discussion

Sources used in this course:

- Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann. ISBN: 978-0-12-381479-1.



- Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). Data Mining: Practical Machine Learning Tools and Techniques (4th ed.). Morgan Kaufmann.
- Tan, P.-N., Steinbach, M., Karpatne, A., & Kumar, V. (2019). Introduction to Data Mining (2nd ed.). Pearson Education.

Evaluation & Grading: 1 midterm exam + 1 final exam + 1 project

Item	Contribution (%)
Midterm	40%
Final	40%
Project	20%

Attendance:

- Regular attendance is expected and important for successfully completing the course.
- Attendance will be monitored throughout the semester in accordance with Ankara University undergraduate education regulations and applicable faculty/departmental rules.

Ankara Üniversitesi Ön Lisans Ve Lisans Eğitim-Öğretim Yönetmeliği

MADDE 38 - (1) Öğrenciler derslere, uygulamalara, öğretim elemanlarınca uygun görülen çalışmalara ve sınavlara katılmak zorundadırlar.

(2) (Değişik: RG-25/10/2018- 30576) Öğrencilerin devam durumu ilgili öğretim elemanı tarafından izlenir ve final sınavından en az bir hafta önce ilan edilir.

Data Mining Term Project (20% of the Course)

The term project is designed to provide students with practical experience in applying data mining techniques to a real-world dataset. Students will define a data mining problem, select and prepare an appropriate dataset, apply multiple data mining algorithms, evaluate and compare the resulting models, and present their findings. Projects will normally be completed in teams of 3-4 students.

Project Proposal – 5%

Each team will submit a short project proposal describing:

- the project title, objective, and problem definition;
- the team members and their responsibilities;
- the selected dataset and its source;
- the intended data mining task (e.g., classification or clustering);
- the planned data mining methods/algorithms; and
- relevant previous studies related to the selected problem.

Project Presentation & Demo – 5%

Each team will present and demonstrate its completed project at the end of the semester. The presentation should briefly introduce the problem and dataset, explain the methodology, and focus on the experimental results and comparison of the selected models.

Students are expected to demonstrate their data mining workflow using WEKA, Orange, Python, or another appropriate data mining environment. All team members are required to participate in the presentation.

Final Project Report – 10%

The final report should document the complete data mining process. At minimum, it should address the following components:

1. Introduction and Related Work

Explain the purpose, motivation, and significance of the project. Discuss relevant previous studies and appropriately cite the literature.

2. Problem Definition and Data Mining Methods

Clearly define the data mining problem and identify whether it involves supervised or unsupervised learning. Students should select and compare at least three appropriate data mining algorithms.

3. Dataset Description

Describe the dataset, its source, size, features/attributes, target or class variable where applicable, and other characteristics relevant to the analysis.

4. Data Preparation and Preprocessing

Explain the preprocessing steps performed before model development. This may include handling missing values and outliers, data transformation, and feature selection. Students should explain and justify important preprocessing decisions rather than simply listing the operations performed.

5. Model Development, Evaluation, and Results

Train and evaluate the selected algorithms using appropriate experimental procedures and performance metrics. Results should be presented clearly, preferably using comparative tables and/or figures. Where applicable, students should compare model performance before and after feature selection.

6. Discussion and Conclusion

Interpret the experimental results and discuss which methods were most appropriate for the selected problem. Findings should be considered in relation to relevant previous studies. Students should also discuss limitations, problems encountered, and possible directions for future work.

References

All datasets, publications, software, and other external resources used in the project must be properly acknowledged. References should follow the IEEE numbered citation style (e.g., [1], [2], [3]).

Note: Detailed instructions concerning submission dates, report format, presentation schedule, dataset approval, and evaluation rubric will be announced separately during the semester from e-kampüs LMS system. Please check the announcements regularly.

Datasets & Repositories- Publicly available datasets

UCI Machine Learning Repository — A long-established repository maintained by the University of California, Irvine for empirical analysis of machine-learning algorithms. It contains datasets suitable for classification, regression, clustering, and other data-mining tasks. https://archive.ics.uci.edu/datasets

Kaggle Datasets — A large platform providing publicly available datasets across many application domains. Students should review the dataset description, provenance, license, variables, and usage restrictions before selecting a dataset for their project.

https://www.kaggle.com/datasets